

Verification status

Repository source: VERIFICATION_STATUS.md

Version: v1.0.0-rc1 **Release date:** unreleased **Status date:** September 19, 2026

Proposed theorem

For every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$, the package represents the proposed universal inequality

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{200000000000000} d^3 = 0.41187756378030227424817892598 \dots d^3.$$

Current designation

Proposed exact universal certificate — standalone internal replay.

The finite exact verifier, independent binary64 Python audit, independent base-R audit, regression tests, adversarial mutation tests, manuscript reconciliation, and documentation checks pass in the recorded environments. The result is not yet externally reproduced, peer reviewed, accepted, tagged, or publicly released.

Candidate publication metadata

- sole named author: Bruce J. Swihart;
- institutional affiliation: none asserted;
- intended repository: <https://github.com/swihart/minvol-degree3-contact-bound>;
- candidate version: v1.0.0-rc1;
- target final tag: v1.0.0;
- prose license: CC BY 4.0;
- software license: MIT;
- final release date and archival identifier: pending.

Verification matrix

ID	Verification item	Evidence	Status
E1	Exact lower-row, fan, determinant, coordinate-width, cap, and handoff inequalities	<code>certificate/certify_universal_bound.py</code> ; certificate JSON; exact transcript	Passed
E2	Byte-identical reconstruction of final machine-readable certificate	Production exact verifier	Passed
N1	Independent binary64 reconstruction	<code>certificate/audit_universal_bound.py</code>	Passed
R1	Independent base-R reconstruction	<code>certificate/audit_universal_bound.R</code> ; archived user-machine transcript	Passed

ID	Verification item	Evidence	Status
T1	Reconstruction and regression suite	Ten Python tests	Passed
M1	Deliberate corruption detection	Seven adversarial mutation tests	Passed
P1	Paper coefficient, split, lower rows, and main constants match certificate	<code>paper/reconcile_certificate.py</code>	Passed
D1	Exact-constants appendix matches certificate	<code>proof/generate_exact_constants.py</code>	Passed
D2	Required release documents and relative links are present	<code>ci/verify_release_docs.py</code>	Passed in this checkpoint
D2a	Candidate author, repository, citation, and split-license metadata are internally consistent	<code>ci/preflight_metadata.sh</code> ; <code>CITATION.cff</code> ; <code>LICENSE</code>	Passed in this checkpoint
D3	Every repository Markdown source has a checksummed PDF counterpart	<code>ci/verify_doc_pairs.py</code>	Passed in construction environment
P2	Paper and documentation PDFs are readable, have embedded fonts, and expose extractable text	Poppler preflight in construction environment and hosted workflow definition	Passed in construction; hosted run pending
C1	Fresh standalone replay with no private path dependency	Clean extraction/bundle tests	Passed for recorded checkpoints
C2	Green hosted Python, R, and document jobs on final public commit/tag	<code>.github/workflows/verification.yml</code>	Pending
X1	Independent mathematical reproduction by an outside reviewer	None yet	Pending
J1	Peer review or publication acceptance	None yet	Pending

What “exact” means here

The theorem-bearing finite comparisons use Python integers and `fractions.Fraction`. The verifier reads immutable integer arrays, checks the semantic inequalities, reconstructs derived artifacts byte for byte, and compares exact rational coefficients. No binary floating-point inequality is used as theorem authority.

“Exact” does not mean that the whole research claim has been independently validated. The proof still depends on the correctness and completeness of the mathematical reduction encoded in the manuscript and verifier, the trusted software/hardware stack, and the absence of a shared conceptual error.

See [certificate/TRUST_BOUNDARY.md](#).

What is established inside the package

- gap-free circumradius coverage from $1/2$ to $\sqrt{6}/4$;
- unique lower-branch bottleneck 03B02;
- exact refinement of the formerly weak lower-row interval;
- final fan reconstruction with 98,306 rays;
- positive exact fan margin;

- determinant transfer and dominant-edge coordinate-width bounds;
- forced-core and three cap-pair lower bounds;
- exact six-cap disjointness;
- a near-Jung lower bound strictly above the bottleneck;
- exact universal coefficient assembly;
- paper and appendix reconciliation with the certificate;
- rejection of seven representative proof-object corruptions.

What is not established

The package does not establish:

- sharpness, equality, or rigidity;
- the identity of a minimizing body;
- the Meissner conjecture;
- external independent reproduction;
- peer review, journal acceptance, or publication;
- an exhaustive priority or novelty determination.

The Meissner volume is used only as an explicit-body comparison value.

Release boundary

Before this release candidate can become an immutable public release, the project still requires:

1. creation of the intended public remote and push of the exact candidate;
2. final author copy edit and approval;
3. green hosted Python, R, and document jobs on that commit;
4. assignment of the final release date and transition to tag `v1.0.0`;
5. a fresh tagged-checkout replay and release-asset hash audit; and
6. explicit approval before the tag, release, or announcement.

External review remains a scientific goal after release and must not be implied by a green workflow.